# Christian González Pérez

Aerospace engineer | Simulation, control and scientific software

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## Profile

Aerospace engineer interested in building tools for engineering problems, combining simulation, control, scientific software and artificial intelligence. Academic and project experience in aerial vehicles, meteorology and space systems. I define, integrate and verify solutions with the support of AI-assisted development tools.

## Education

**Bachelor's degree in Aerospace Engineering** 2026

University of León, Spain | Average grade: 9.17/10. Extraordinary End-of-Degree Award, 2025–2026 academic year.
Undergraduate thesis: cascaded hybrid quadcopter control with neural replacement of the outer loop. Grade: 9.5/10.

**University microcredential in Model-Based Engineering** 2026

University of León | 5 ECTS credits. Systems modelling with Capella.

**Advanced course in simulation (Simulación Aérea)** 2025–2026

University of León | 6 credits.

## Technical projects

**Neural control of a quadcopter – undergraduate thesis** 2026

Repository | Project and thesis

- Designed a modular six-degree-of-freedom simulator in Python to compare classical controllers and MLP, GRU and LSTM networks for trajectory tracking.
- Defined the physics and control architecture and evaluated tracking on known and unseen trajectories. Neural networks achieved comparable or better results in some cases, with no overall superiority. The study was conducted entirely in simulation.

**SkyBrief – aeronautical weather** 2026

Repository

- Developed an academic prototype for retrieving METAR and TAF reports, weather maps and route profiles for over 700 European airports.
- Integrated clearly separated current reports and historical ERA5 episodes, with weather risk indicators. This is a prototype, not an operational tool.

## Technical projects (continued)

**FireSat25 / ULEsat25 – CubeSat mission** 2025–2026

- Contributed to the academic design of a wildfire observation mission as part of a four-person onboard data handling (OBDH) team.
- The team proposed an onboard computer architecture and built a Python client-server demonstrator for simulated telemetry and telecommands. I presented the FireSat25 mission at a poster session in 2025.

## Leadership and other activities

**Founder and president, Club de Moda ULe** 2024–2026

Coordinated a university association of around 15 active members and liaised with the university and external collaborators. Organised a fashion show with the team, involving around 50 attendees, 10 models and four retailers. Developed interactive web applications for association activities.

**Competitions and team projects** 2024–2026

- D.R.A.G.: a conceptual passive aerodynamic braking project for space vehicles, one of the winners of the 2026 ULE Prototype Competition; team project with preliminary simulations and no physical prototype.
- Second place with team R.A.R. at EUDIS Defence Hackathon Spring 2026 Spain. Participated in NASA Space Apps Challenge León 2024.

## Skills and languages

**Engineering:** flight dynamics simulation and control; aeronautical meteorology; model-based engineering; data analysis and visualisation.
**Software:** Python, PyTorch, Git and GitHub; web application experience with JavaScript, HTML, CSS and React. Use of AI coding agents with requirements definition and result verification.
**Languages:** native Spanish; English: Cambridge B2 (2022); comfortable in technical conversations.
**Mobility:** willing to relocate within Spain or internationally.